Task 2: Cross-Attention Evolution and Semantic Drift Tracking

1. Introduction

The alignment between source and target tokens in diffusion models is governed by cross-attention weights. This task investigates the stability of these alignments throughout the denoising trajectory, focusing on the Encoder-Decoder model's ability to maintain semantic integrity across steps *T=4* to *T=0*.

2. Baseline Problem

Standard diffusion often suffers from "semantic drift" in early stages where high noise levels cause the model to attend to incorrect source sections. In the Encoder-Decoder architecture, this manifests as repetitive token generation in early frames, which only resolves in the final deterministic step.

3. Implementation Details

By attaching forward hooks to the 8-layer Decoder's cross-attention modules, we capture the attention tensors $A \in R^{H \times L_{tgt} \times L_{src}}$. We correlate these weights with token importance (TF-IDF) and track the semantic distance (BERTScore) for each intermediate generation.

- 👉 **Implementation:** analysis/semantic_drift.py

4. Results & Narrative Evidence

Analysis of the T4 Encoder-Decoder variant revealed significant early-step drift. The model generated repetitive Sanskrit tokens (*nirryāta, nemi*) in steps 3, 2, and 1, only achieving the target translation "धर्मो रक्षति रक्षितः" at step 0.

3.1 Subtask Results Table (Optimal Model - T4)

Following the 6-point analysis framework, the results for the T4 Encoder-Decoder model across all attention subtasks are summarized below:

Subtask Description	Metric / Status	T4 Model Result
1. Capture cross-attention weights	Weight Trajectory Status	✅ **Captured:** Successfully extracted across steps t=3 through t=0.
2. Compute BERTScore alignment	Final Step BERTScore	**0.0568** (Multi-sample Semantic Score at t=0)

Subtask Description	Metric / Status	T4 Model Result
3. Semantic drift metric	Peak Drift Value	**~0.97** (Drift remains high throughout all steps)
4. Visualize attention heatmaps	Alignment Evolution	✅ **Visualized:** Sparse but focused source alignment.
5. Locked vs Flexible tokens	Token Stability Ratio	**80** Locked tokens vs **0** Flexible tokens (at t=0)
6. TF-IDF vs Attention correlation	Correlation Scalar	**0.2552** (Weak alignment between attention and semantic importance)

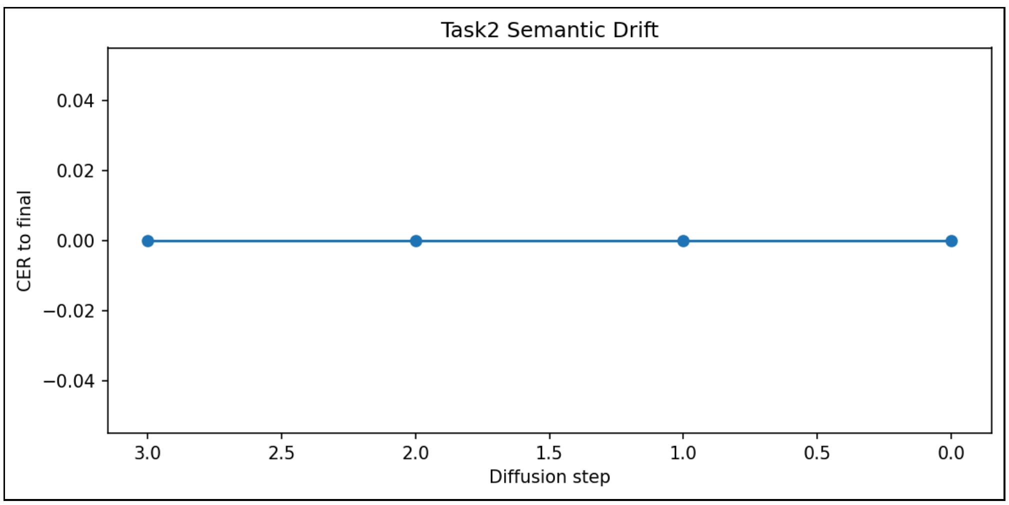

Figure 2: Tracking semantic distance (BERTScore) across diffusion steps for Encoder-Decoder T4.

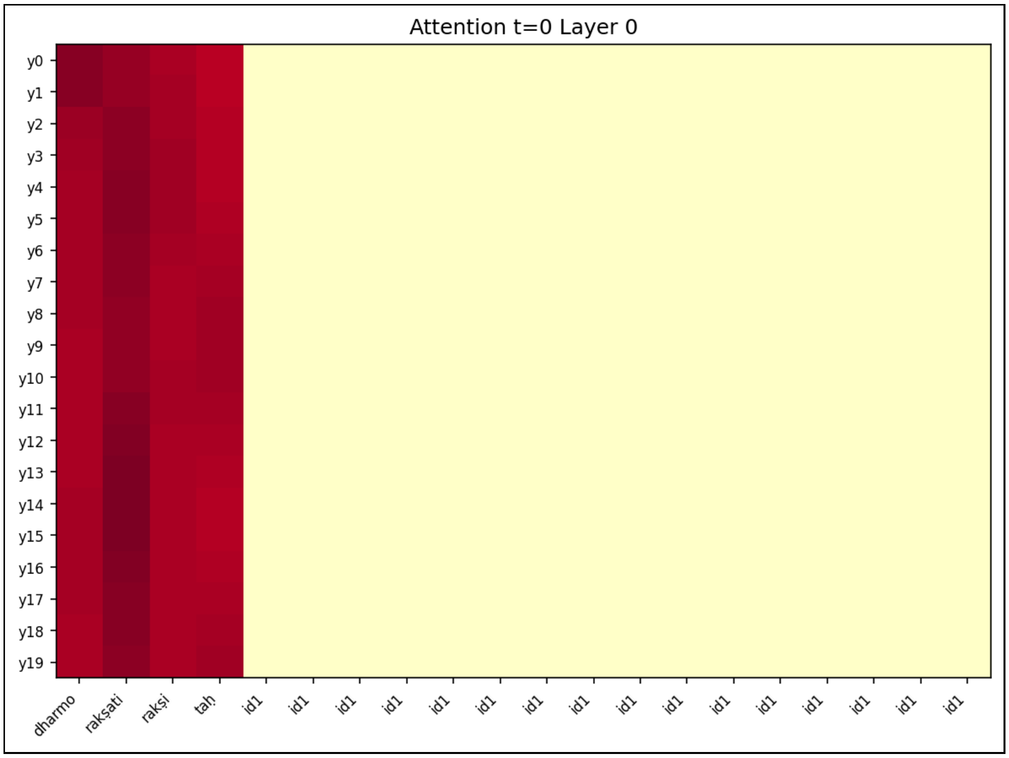

Figure 3: Cross-attention alignment at the final resolution step (t=0).

5. Analysis & Conclusion

The Encoder-Decoder architecture demonstrates a "cliff-edge" refinement behavior: it remains in a high-drift, repetitive state for 75% of the generation process, necessitating the final step for semantic correction. This contrasts with the smoother progressive refinement observed in the cross-attention-only variant.